AJAYI, SUNDAY IYANU

(+250) 793 150 023 | sajayi@andrew.cmu.edu | sunnexajayi@gmail.com | LinkedIn |Github

SUMMARY

Sunday is a Software Engineer and master's student in Information Technology (Applied Machine Learning) at Carnegie Mellon University Africa, with over four years of experience building scalable, data-driven systems across fintech, transportation, and digital infrastructure. His research interests span applied machine learning for intelligent energy and renewable power systems, digital accessibility and financial inclusion, and AI safety, governance, and policy frameworks for the responsible development and deployment of AI across Africa.

EDUCATION

Carnegie Mellon University, Africa **Expected Graduation:** May, 2027
Master of Science in Information Technology
- **Concentration:** Applied Machine Learning and Systems
- **Relevant Coursework:** Machine Learning, Software Architecture, Recommender Systems, Advanced Algorithms

Obafemi Awolowo University, Nigeria **Graduated:** April, 2018
B.Sc in Electronic and Electrical Engineering
- **Specialization:** Control Systems, System Design & Architecture

RESEARCH EXPERIENCE

Graduate Researcher 2025 – Present
Carnegie Mellon University Africa
- Designed a voice-driven USSD automation framework integrating automatic speech recognition (ASR), biometric authentication, and conversational interaction workflows.
- Conducted security, accessibility, and usability evaluations for visually impaired users in mobile money scenarios.
- Investigated deployment feasibility on low-cost mobile devices and low-connectivity environments.

PUBLICATIONS

Ajayi, S. I., et al. (Under Review) 2026
Toward Accessible Mobile Money: A Voice-Driven, Biometrically Secured USSD Automation Framework for Visually Impaired Users.

TECHNICAL SKILLS

Languages & Backend: Python, Go, Django, Flask | **Applied ML & Data Tools:** Pytorch, hugging face, Numpy, Pandas, ASR/TTS, LLM Applications, BeautifulSoup, Celery | **Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis | **Systems Architecture:** Microservices Architecture, API Gateway, JWT Authentication, Caching Strategies, Queuing | **Infrastructure & DevOps:** AWS, GCP, Azure, Docker, CI/CD, Nginx, Prometheus, Grafana, Git | **Communication & Testing:** REST, gRPC, Webhooks, PyTest, Postman, Swagger.

PROFESSIONAL EXPERIENCE

Backend Engineer (Go | Python) Mar 2024 – Sep 2025
Mercurie, Lagos
- Built and optimized backend systems using gRPC and REST, reducing latency by 30% and improving feature deployment speed by 40%.
- Maintained 99.9% uptime for high-traffic SaaS products.

Backend Engineer (Python) Feb 2023 – Feb 2024
Jenesys HQ, United Kingdom
- Architected and developed a high-performance backend achieving 40% faster response times and zero downtime during peak load.
- Integrated Sage, Xero, Slack, and WhatsApp to reduce management overhead by 30% and improve user engagement by 25%.

Backend Engineer (Python) Sep 2022 – Apr 2023
PressOne Africa, Lagos, Nigeria
- Enhanced a telephony backend handling 20,000+ API requests and 5,000+ call events monthly with 95% uptime.
- Delivered tools improving issue resolution by 40% and integrated Telynx and Plivo to cut onboarding time by 60%.

Backend Engineer (Python) Nov 2019 – July 2023
SecureID Digital, Lagos, Nigeria
- Led the backend architectural design and deployment of scalable systems, improving uptime by 40% and reducing incident response time by 60%
- Served as Technical Product Lead and Lead Backend Engineer for Nigeria Railway's E-ticketing system project, supporting 300,000+ users and 5,000+ daily transactions.

CERTIFICATIONS

- Google Professional Cloud Architect 2024
- AWS Cloud Practitioner 2021

EXTRACURRICULAR ACTIVITIES AND INTERESTS

Leadership: Founder, YE Network | Minister of Partnership, Africa Innovators Society, CMU- Africa | Community Lead, Python Rwanda, AWS Community Builder
Membership: Software Engineering & Cybersecurity Society, CMU Africa, Data Science Club, IEEE – CMU Africa,
Interests: Mentor Youths for social impact | Mentor Software engineer | Reading | IoT DIY Projects | Volunteering